ASSESSMENT TOOL CONSTRUCT – Class 9 & 10

PART 1 — RIASEC STRUCTURE

Holland's 6 traits : Total 24 questions

Code	Meaning	No of Questions
R	Realistic	4
I	Investigative	4
A	Artistic	4
S	Social	4
E	Enterprising	4
C	Conventional	4

Scoring Method : 5-point scale

Response	Score
Strongly Disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly Agree	5

RIASEC Score Calculation Method

Say for Enterprising questions student scores : 4 + 5 + 4 + 3 = 16

Maximum possible : 20

Then Enterprising Score = 16/20 X 100 = 80%

Similarly we will get scores for all the 6 areas – Realistic, Investigative, Artistic, Social, Enterprising, Conventional.

Now classify each of the 6 traits as :

Score	Grading Level	Level Meaning
75 – 100	Highly Preferred	A strong trait with natural interest, you can expertise
60 – 74	Fairly Good	Meaningful interest, above the crowd, can be developed further
45 – 59	Not Decisive	Not a prominent interest but it's there
Below 45	Best Avoided	Activities associated with this trait may be less naturally appealing

Tie-break & Edge case Rules

Step 1 - When 2+ trait scores are equal upto 2 decimal places, Higher-ranking trait = fewer neutral (score=3) responses among its 4 questions

Step 2 - Still tied after Step 1, apply fixed fallback priority order: I > E > S > A > C > R

To Note :

Comment 1 - All 6 RIASEC traits score below 45% ("Best Avoided" band), attach flag: "No Strong RIASEC Preference Emerging – recommend qualitative follow-up"

Comment 2 - Max score minus min score across all 6 traits is less than 15 points, attach flag: "Highly Undifferentiated Profile"

PART 2 — BIG FIVE

Big Five dimension : Total 20 questions

Trait	Meaning	No of Questions
Openness	Creativity & curiosity	4
Conscientiousness	Discipline	4
Extraversion	Social energy	4
Agreeableness	Empathy	4
Emotional Stability	Stress handling	4

Positively Keyed + Reverse Keyed Used

Example :

Positive : "I enjoy meeting new people."

Reverse : "I avoid social interactions whenever possible."

Reverse Scoring Formula : If response = 5 (Strongly Agree)

Reverse score becomes, Reverse Score = 6 – Original Response = 6 - 5 = 1

This prevents fake answering. 8 such questions are there in the Class 9 & 10 questionnaire.

Big Five Score Calculation Method

Say for Openness questions student scores : 4 + 5 + 4 + 3 = 16 Maximum possible : 20

Then Openness Score = 16/20 X 100 = 80%

Similarly we will get scores for all the 5 traits – Openness, Conscientiousness, Extraversion, Agreeableness, Emotional Stability.

Now classify each trait as :

Score	Grading Level	Level Meaning
75 – 100	Highly Evident	This personality tendency is strongly visible in your behaviour across situations.
60 – 74	Evident	This tendency is generally present and influences your behaviour in many situations.
45 – 59	Moderately Evident	This tendency appears occasionally but is not a defining characteristic.
Below 45	Less Evident	This tendency is not naturally prominent in your behavioural style.

Kindly Note :

1. When calculating the percentage for each trait, reverse-keyed questions use the reverse-converted score (e.g. 1, not 5).
2. When calculating the Response Validity Score, reverse-keyed questions use the student's original punched response (e.g. 5, not 1).

Tie-break & Edge case Rules

Step 1 - When 2+ trait scores are equal upto 2 decimal places, Higher-ranking trait = fewer neutral (score=3) responses among its 4 questions

Step 2 - Still tied after Step 1, apply fixed fallback priority order: O > C > E > A > S

To Note :

Comment 1 - All 6 BIG Five traits score below 45% ("Best Avoided" band), attach flag: " Stress Vulnerability – Flag for Counsellor Attention"

Comment 2 - Rank 1 and Rank 2 scores are within 5 points of each other, attach flag: " Balanced Personality Profile, Low Differentiation"

PART 3 — APTITUDE

Aptitude dimension : Total 20 questions, 15 min (45 sec for each question)

Area	Measures	No of Questions
Numerical Reasoning	Numbers, calculations, quantitative thinking	5
Verbal Reasoning	Language comprehension & interpretation	5
Logical Reasoning	Pattern detection & analytical logic	5
Spatial Reasoning	Visualization & mental rotation	5

Difficulty Level for each Area	% split among difficulty	No of Questions	Scoring Weightage
Easy	40%	2	1
Medium	40%	2	1.5
Hard	20%	1	2

Aptitude Score Calculation Method

Example for Numerical Reasoning, lets assume this is how we made the questions.

Question	Difficulty	Weight
Q1	Easy	1
Q2	Easy	1
Q3	Medium	1.5
Q4	Medium	1.5
Q5	Hard	2

Maximum possible score : 1 + 1 +1.5 + 1.5 + 2 = 7 Student scores = say 4.5

Then Numerical Reasoning Score = 4.5/7 X 100 = 64.3%

Similarly we will get scores for all the 4 traits – Numerical Reasoning, Verbal Reasoning, Logical Reasoning & Spatial Reasoning.

Now classify each trait as :

Score	Grading Level	Level Meaning
75 – 100	Advanced Capability	Demonstrates strong natural potential in this area and can perform complex tasks with relative ease.
60 – 74	Good Capability	Shows above average potential and can perform effectively with practice and experience.
45 – 59	Developing Capability	Demonstrates moderate potential but may require additional training and exposure.
Below 45	Needs Development	This area may require significant effort, practice and support to achieve strong performance.

<u>Tie-break & Edge case Rules</u>

Step 1 - When 2+ trait scores are equal upto 2 decimal places, Higher-ranking trait = the one with the higher Difficulty Consistency (DC) score for that trait (from the ARI calculation)

Step 2 - Still tied after Step 1, apply fixed fallback priority order: Logical Reasoning > Numerical Reasoning > Verbal Reasoning > Spatial Reasoning

PART 4 — COGNITIVE & DECISION STYLE

Cognitive & Decision dimensions : Total 9 questions

Trait	Purpose	No of Questions
Learning Velocity	Adaptability / fixed mindset	3
Uncertainty Tolerance	Risk comfort / entrepreneurship readiness	3
Autonomy Preference	Independent / team working	3

Scoring Method : 5-point scale

Response	Score
Strongly Disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly Agree	5

<u>C&D Score Calculation Method</u>

Say student scores : 4 + 5 + 3 = 12 Maximum possible for each Trait : 15

Then Trait Score = 12/15 X 100 = 80%

Similarly we will get scores for all the 3 areas – Learning Velocity, Uncertainty Tolerance & Autonomy Preference.

Now classify each area as :

Score	Grading Level	Level Meaning
75 – 100	Strongly Demonstrated	This behaviour is consistently visible and is likely to support workplace effectiveness.
60 – 74	Demonstrated	This behaviour is generally present but may vary across situations.
45 – 59	Emerging	This behaviour appears occasionally but is not yet consistently demonstrated.
Below 45	Needs Development	This behaviour is currently less evident and may require conscious development.

Tie-break & Edge case Rules

Step 1 - When 2+ trait scores are equal upto 2 decimal places, Higher-ranking trait = fewer neutral (score=3) responses among its 3 questions

Step 2 - Still tied after Step 1, apply fixed fallback priority order: Learning Velocity > Uncertainty Tolerance > Autonomy Preference

APTITUDE RELIABILITY INDEX (ARI)

ARI Components :

Component	Purpose	Weightage
Difficulty Consistency - DC	Detect abnormal answer pattern - guesswork	0.6
Time Consistency - TC	Detect random clicking & disengagement	0.4

STEP 1 — DIFFICULTY CONSISTENCY

For each trait, the student responses as below which is statistically unusual :

Question	Difficulty	Possibility 1	Possibility 2	Possibility 3	Possibility 4	Possibility 5	Possibility 6
Q1	Easy	✗	✗	☑	✗	✗	✗
Q2	Easy	✗	✗	✗	☑	✗	✗
Q3	Medium	✗	☑	✗	✗	☑	☑
Q4	Medium	✗	✗	✗	☑	✗	☑
Q5	Hard	☑	☑	☑	☑	✗	✗

Scoring = 100 + Apply penalties.

Penalty Rules

Possibility No	Condition	Penalty
1	Hard correct + all other wrong	-25
2	all Easy wrong + any Medium correct + Hard correct	-20
3	any Easy correct + all Medium wrong + Hard correct	-15
4	any Easy correct + any Medium correct + Hard correct	-10
5	all Easy wrong + any Medium correct + Hard wrong	-15
6	all Easy wrong + all Medium correct + Hard wrong	-15
	All other possibilities	0

Student scored as Hard correct + all other wrong (Possibility No 1)

Then DC = 100 – 25 = 75

Calculate for each question of the trait. Then the final score is the average of the 5 responses of the trait. The average of the 4 trait score is the final sore.

STEP 2 — TIME CONSISTENCY

Expected time to complete each question = 45 sec

If the student has answered hard questions in 3–5 seconds & yet correct, statistical possibility of guessing is high.

Time Consistency Rules

Condition	Penalty
Easy & Medium question answered within 5 sec & wrong	-5
Hard question answered within 5 seconds (correct or wrong & it's not 'Not Sure')	-10

A student answers a hard question in 4 sec and correct or wrong.

Penalty : – 10

TC for that specific question = 100 – 10 = 90

Calculate for each question of the trait. Then the final score is the average of the 5 responses of the trait. The average of the 4 trait score is the final sore.

By looking at the FINAL ARI Score, one can say how confident the student appears for his aptitude, also known as Assessment Confidence.

ARI = (DC × 0.6) + (TC × 0.4)

Final ARI range : 0 – 100

ARI	Grading Level	Level Meaning
75–100	High Reliability	Results are trustworthy and can be used confidently for career recommendations.
60–74	Moderate Reliability	Results are generally reliable but should be interpreted alongside other assessment findings.
Below 60	Low Reliability	Results may be affected by guessing, disengagement or inconsistent responding and should be reviewed carefully.

APTITUDE CONFIDENCE INDEX (ACI)

ACI = (No of **"Don't Know / Not Sure" / Total Questions) X 100 %**

DK % Score	No of Q	Grading Level	Level Meaning
0 – 15	3	High Confidence	Demonstrates strong willingness to engage with unfamiliar or challenging problems.
16 – 30	6	Balanced Confidence	Shows a healthy balance between attempting questions and recognizing uncertainty.
31 – 45	9	Developing Confidence	Tends to hesitate when unsure and may benefit from confidence-building experiences.
Above 45		Limited Confidence	Frequently avoids uncertain situations, indicating a need for greater confidence and problem-solving exposure.

RESPONSE VALIDITY SCORE (RVS)

This applies to RIASEC, Big Five & Cognitive and Decision Style, NOT Aptitude.

COMPONENT — INTERNAL CONSISTENCY : Score Range 0 - 100

Question 1 : "I enjoy meeting new people."

Question 14 : "I avoid social interactions whenever possible."

These are opposite constructs. Expected response would be Q1 – Agree / Strongly Agree Q2 – Disagree / Strongly Disagree

Instead if response is Agree / Strongly Agree for both questions, then contradiction detected. Based on the rating scale, measure the gap between the ratings.

Gap	Meaning
3–4	Good consistency
2	Acceptable
1	Mild contradiction
0	Strong contradiction

Now map the penalties as below :

Contradiction Type	Calculation	Penalty
Mild contradiction	To be calculated for each pair	-5
Strong contradiction	To be calculated for each pair	-10
Multiple contradictions	Upto 4 contradictions	-15
Severe inconsistency pattern	5 – 10 contradictions	-25

For effective measurement at this category, we have 10 mirror pairs.

Kindly note that in the case of any mirror pair, if either of the question is a reverse key, then for calculating the RVS, consider the response for calculation. For eg in Q36, if students punches 2 as his response, then for calculating RVS we need to consider 2. For calculating the percentage of the trait, we need to consider 6-2 = 4

For all the 10 mirror pairs, calculate the RVS score individually. Then the average of the result will be the final score. Finally after score compilation, arrive at the grading as below :

Score	Grading Level	Level Meaning
80–100	Highly Consistent Profile	Responses demonstrated strong internal consistency and a clear understanding of personal tendencies.
60–79	Generally Consistent Profile	Responses were mostly consistent, though a few contradictions suggest some uncertainty or situational variation.
Below 60	Inconsistent Profile	Responses contained multiple contradictions, requiring cautious interpretation of some assessment findings.

OVERALL ASSESSMENT RELIABILITY INDEX (ORI)

Method : Note the total completion time for the assessment. Then grade the time taken with the benchmark as below.

Completion Time	Grading Level	Level Meaning
27 – 36 min	High Reliability	Ideal completion behaviour
22 – 26 min	Moderate Reliability	Slightly fast
37 – 45 min	Moderate Reliability	Slightly slow
16 – 21 min	Low Reliability	Likely rushed
46 – 60 min	Low Reliability	Excessively slow
Below 16 min	Very Low Reliability	Highly unreliable
Above 60 min	Very Low Reliability	Strong validity concern

DOMINANT CAREER STYLE (DCS)

Method : For all the possible 3 letter RIASEC code combinations mentioned in the Traits&Weightages file (sheet – RIASEC 120), a dominant style is mentioned in the excel sheet. Against that a description is given. Now against each 3 letter code, check the score of the student. For eg for code RIA, check cumulatively what the student got for Realistic, Investigative & Artistic trait. Likewise for all 120 codes. Then take the top one. Tie-break and edge case rules mentioned in the excel sheet.

DOMINANT PERSONALITY STYLE (DPS)

Method : For all the possible 2 letter BIG Five combinations mentioned in the Traits&Weightages file (sheet – BIG Five 20), a dominant style is mentioned in the excel sheet. Against that a description is given. Now against each 2 letter code, check the score of the student. For eg for code O-C, check cumulatively what the student got for Openness & Conscientiousness trait. Likewise for all 20 codes. Then take the top one. Tie-break and edge case rules mentioned in the excel sheet.

STREAM RECOMMENDATION ENGINE

All streams possible are as below :

Main Stream	Sub-Streams	Core Subjects Usually Offered
Science	PCM (Physics, Chemistry, Maths)	Physics, Chemistry, Mathematics, English
Science	PCB (Physics, Chemistry, Biology)	Physics, Chemistry, Biology, English
Science	PCMB	Physics, Chemistry, Mathematics, Biology
Science	PCM + Computer Science	Physics, Chemistry, Mathematics, Computer Science
Science	PCM + Economics	Physics, Chemistry, Mathematics, Economics

Science	PCB + Psychology	Physics, Chemistry, Biology, Psychology
Commerce	Commerce with Maths	Accountancy, Business Studies, Economics, Mathematics, English
Commerce	Commerce without Maths	Accountancy, Business Studies, Economics, English
Commerce	Commerce + Informatics Practices	Accountancy, Business Studies, Economics, IP
Commerce	Commerce + Computer Science	Accountancy, Economics, CS
Humanities / Arts	Humanities with Economics	History, Political Science, Economics, English
Humanities / Arts	Humanities with Psychology	Psychology, Sociology, Political Science, English
Humanities / Arts	Humanities with Maths	Geography, Economics, Mathematics, English
Humanities / Arts	Humanities with Legal Studies	Political Science, Legal Studies, History
Humanities / Arts	Humanities with Fine Arts	Fine Arts, History, English
Humanities / Arts	Humanities with Mass Media	Political Science, Sociology, English

For each stream, top 5 traits out of the 18 are identified which are important for that specific stream. A weighted score is given for the top 5 traits. The total weightage will be 100%.

The student will have his score as well as rating against each of the 18 traits. His equivalent trait score needs to be considered for each stream. Then arrive at the FIT Score for each of the stream selected. Then select the top 3 streams as suggested by the Fit Score.

Example, Sub-Stream = Commerce with Maths

Trait	Student Fit Score	Weight
Enterprising	90	30
Verbal	80	20
Extraversion	75	25
Conscientiousness	70	10
Uncertainty Tolerance	50	15

For the Stream selected, Fit Score = ∑ ("Trait Score" X "Weightage")

Final Stream Score = (90X0.3)+(80X0.2)+(75X0.25)+(70X0.1)+(50X0.15) = 75 %

Top 3 Sub-Streams from the Main Stream can be selected after complete ranking. The Stream Fit Score shall also have a grading as below :

Score	Grading Level	Level Meaning
75 – 100	Strong Fit	Strong point
60 – 74	Good Fit	With skill training, will do well
Below 60	Weak Fit	Consider with due diligence

CAREER MATCHING FORMULA

There are now 18 traits as below for which ratings have been arrived for the student.

Layer	Trait

RIASEC	Realistic
RIASEC	Investigative
RIASEC	Artistic
RIASEC	Social
RIASEC	Enterprising
RIASEC	Conventional
Big FIVE	Openness
Big FIVE	Conscientiousness
Big FIVE	Extraversion
Big FIVE	Agreeableness
Big FIVE	Emotional Stability
Aptitude	Numerical Reasoning
Aptitude	Verbal Reasoning
Aptitude	Logical Reasoning
Aptitude	Spatial Reasoning
Cognitive & Decision	Learning Velocity
Cognitive & Decision	Uncertainty Tolerance
Cognitive & Decision	Autonomy Preference

For each domain in the career library, top 5 traits out of these 18 are identified which are important for that specific career domain. A weighted score is given for the top 5 traits. The total weightage will be 100%.

In case of a tie, the trait with higher average response consistency (fewer neutral answers) ranks higher.

The student will have his score as well as rating against each of these 18 parameters. His equivalent trait score needs to be considered for each Domain. Then arrive at the FIT Score for each of the Domain selected. Then select the top 6 Domain as suggested by the Fit Score. Take the first career from each of these domains for session 1 purpose.

Example, Domain = Investment Banking

Trait	**Student Fit Score**	**Weight**
Enterprising	80	35%
Numerical	80	20%
Extraversion	60	15%
Conscientiousness	60	15%
Verbal	70	10%

Career Fit Score = ∑ ("Trait Score" X "Weightage")

Career Fit Score = (80 X 0.35)+(80X0.2)+(60X0.15)+(60X0.15)+(70X0.10) = 69%

Top 6 careers can be selected after complete ranking. The Career Fit Score shall also have a grading as below :

Score	**Grading Level**	**Level Meaning**
75 – 100	Strong Fit	Strong point
60 – 74	Good Fit	With skill training, will do well
Below 60	Weak Fit	Consider with due diligence